

A Disproof of Conjecture 00000001299: Ducci Cycle Lengths Need Not Be Odd Factors of n

earthking11

September 13, 2026

Abstract

Conjecture 00000001299 asserts that for n a power of two the Ducci sequence reaches zero, that for non-powers of two nontrivial cycles exist, *and that the minimal cycle length is an odd factor of n* . We refute the last clause. For $n = 5$ the seed $(0, 0, 0, 1, 1)$ lies on a cycle of minimal non-trivial length 15; the number 15 is odd but $15 \nmid 5$. In particular the minimal cycle length is *not* an odd factor of n . For $n = 10$ the minimal non-trivial period is again 15, and $15 \nmid 10$. The first clause (the classical Ducci zero criterion) is not in dispute and is stated here only for context. The counterexample is verified by direct iteration, formalized in core Lean 4, and reproduced by an independent Python script.

1 The Ducci map and the conjecture

Fix an integer $n \geq 2$. The *Ducci map* acts on integer n -tuples by taking cyclic absolute differences,

$$\mathsf{T}(x_0, x_1, \dots, x_{n-1})_i = |x_i - x_{i+1}|, \quad i \in \mathbb{Z}/n\mathbb{Z}, \quad (1)$$

and a *Ducci sequence* is the orbit $x, \mathsf{T}x, \mathsf{T}^2x, \dots$ of some seed. The conjecture file `00000001299.md` states (verbatim):

“For n a power of two the sequence reaches zero in finitely many steps, while for non-powers of two nontrivial cycles exist; and the minimal cycle length is an odd factor of n .”

(The file also gives an equivalent Chinese statement. We quote the English version, which is the operative wording.)

The classical part of this statement is the *Ducci zero criterion*: the sequence reaches $(0, \dots, 0)$ for every seed *iff* n is a power of two; for n not a power of two nonzero cycles exist. That part is true and is *not* refuted below. Our counterexample concerns only the final clause, “the minimal cycle length is an odd factor of n .”

2 The counterexample: a 15-cycle for $n = 5$

Take $n = 5$ and the seed

$$s = (0, 0, 0, 1, 1).$$

Iterating (1), the orbit of s is

$T^0 s = (0, 0, 0, 1, 1)$	$T^1 s = (0, 0, 1, 0, 1)$	$T^2 s = (0, 1, 1, 1, 1)$
$T^3 s = (1, 0, 0, 0, 1)$	$T^4 s = (1, 0, 0, 1, 0)$	$T^5 s = (1, 0, 1, 1, 1)$
$T^6 s = (1, 1, 0, 0, 0)$	$T^7 s = (0, 1, 0, 0, 1)$	$T^8 s = (1, 1, 0, 1, 1)$
$T^9 s = (0, 1, 1, 0, 0)$	$T^{10} s = (1, 0, 1, 0, 0)$	$T^{11} s = (1, 1, 1, 0, 1)$
$T^{12} s = (0, 0, 1, 1, 0)$	$T^{13} s = (0, 1, 0, 1, 0)$	$T^{14} s = (1, 1, 1, 1, 0)$

The 15 displayed states are pairwise distinct, and closing the cycle,

$$T^{15} s = T(1, 1, 1, 1, 0) = (0, 0, 0, 1, 1) = s.$$

Hence s lies on a cycle of length 15. Since no $T^k s$ with $1 \leq k \leq 14$ equals s , the *minimal* non-trivial period of s is exactly 15.

Theorem 1. *The minimal non-trivial Ducci cycle length for $n = 5$ is not an odd factor of n .*

Proof. For the seed $s = (0, 0, 0, 1, 1)$ the orbit is periodic with minimal non-trivial period 15, as just computed. The integer 15 is odd, but it is larger than $n = 5$, so it cannot divide 5: indeed a divisor of 5 is at most 5. Thus the minimal non-trivial cycle length 15 is not an odd factor of $n = 5$, contradicting the final clause of the conjecture. $\square$

3 Exhaustive and randomized period data

The claim that 15 is the minimal non-trivial period for $n = 5$ is not an artifact of one seed. Table 1 reports the complete set of eventual periods observed by exhaustive enumeration and by random sampling. For $n = 5$, exhaustive iteration over all 10^5 seeds in $\{0, \dots, 9\}^5$ yields the period set $\{1, 15\}$; the value 1 is the trivial fixed point $(0, \dots, 0)$, so the minimal non-trivial period is 15. The same minimal period 15 occurs for $n = 10$, where exhaustive iteration over all $4^{10} = 1,048,576$ seeds in $\{0, \dots, 3\}^{10}$ yields the period set $\{1, 15, 30\}$.

Table 1: Eventual Ducci periods. “min. non-triv.” omits the trivial fixed point of period 1. Every listed minimal non-trivial period is odd; none of the ones in boldface divides its n .

n	seeds tested	period set	min. non-triv.
5	exhaustive $\{0, \dots, 9\}^5$ (10^5)	$\{1, 15\}$	15
5	random, values $\leq 10^6$ (2000)	$\{15\}$	15
10	exhaustive $\{0, \dots, 3\}^{10}$ (4^{10})	$\{1, 15, 30\}$	15
10	random, values $\leq 10^6$ (300)	$\{15, 30\}$	15
11	exhaustive $\{0, 1, 2\}^{11}$ (3^{11})	$\{1, 341\}$	341
13	random, values ≤ 50 (20000)	$\{819\}$	819

In every case the minimal non-trivial period fails to divide n :

$$15 \nmid 5, \quad 15 \nmid 10, \quad 341 \nmid 11, \quad 819 \nmid 13.$$

For $n = 5$ and $n = 10$ this is immediate because the period exceeds n ; in fact $15 > 5$ and $15 > 10$.

4 The classical zero criterion (context, not disputed)

The first two clauses of the conjecture are a well-known theorem: every Ducci sequence reaches the zero tuple exactly when n is a power of two, and for every other n there exists a seed whose orbit is a nontrivial cycle. Our $n = 5$ and $n = 10$ examples are non-powers of two, consistent with that criterion; the refutation is directed solely at the assertion that the *minimal* cycle length is an odd factor of n . In other words, the conjecture conflates a correct existence statement (nontrivial cycles for non-powers of two) with an incorrect arithmetic description of their lengths.

Conjecture 2 (00000001299, refuted). For non-powers of two the minimal Ducci cycle length is an odd factor of n .

5 Honest notes and caveats

Remark 3 (A non-rigorous argument we do *not* use). An earlier attempt at a proof ran as follows: “the mod-2 reduction of the Ducci map on $n = 5$ has periods $\{1, 15\}$, hence every integer cycle length must be divisible by 15; in particular $15 \nmid 5$ refutes the conjecture.” This argument is **not** rigorous and is not used here. The reduction modulo 2 does not control the period of the integer orbit: in one sample of 300 random seeds for $n = 5$, 16 seeds have integer period 15 while their mod-2 reduction has period 1 (the count is random-sample dependent; the phenomenon is robust). Reductions of a map can merge or collapse orbits, so periods of a reduction need not divide or be divided by periods in the original system. Our conclusion rests instead on direct iteration of (1), which is finite and exact (and is formalized below).

Remark 4 (The strongest objection: a vacuous reading). A sceptic could try to rescue the conjecture by reading “minimal cycle length” as including the trivial fixed point $(0, \dots, 0)$ of length 1. Then 1 is an odd factor of every n , and the final clause becomes vacuous for all n — which cannot be the intended content, since the conjecture explicitly contrasts “nontrivial cycles” with the zero tuple. Under the natural reading (the minimal length of a *nontrivial* cycle), the clause is false at $n = 5$. We note that the $n = 9$ case happens to satisfy the clause (its minimal non-trivial period is $3 = 9/3$), but a single counterexample at $n = 5$ is enough to refute a universal claim; the clause is simply not a valid general description of Ducci cycle lengths.

Remark 5 (A possible garbling: $2^k - 1$ rather than “factor of n ”). The number $15 = 2^4 - 1$ is not a factor of 5, but it *does* fit a different familiar pattern: for $n = 5$ the minimal non-trivial period is $2^{\text{ord}} - 1$ with $\text{ord} = 4$. It is plausible that “an odd factor of n ” is a garbling of “an odd divisor of $2^k - 1$ for some k ,” or of the statement that Ducci periods are odd. But neither reading is the literal wording of the conjecture file, and neither rescues it: 15 is not an odd factor of 5, which is exactly what the file claims. We therefore record the verdict as **FALSE** under the stated wording, while flagging that the intended mathematical content may have been something else.

6 Formalization and reproduction

The counterexample is machine-checked in core Lean 4 (toolchain `leanprover/lean4:v4.33.1`, `import Std only`; no `Mathlib`, no `sorry`, no `axiom`, no `native_decide`). The file `lean4/Main.lean` defines `T` on `Fin 5 → Nat` via truncated subtraction, the seed `s`, and the iterate `iter`; it proves that `iter 15 = s` and that no positive iterate below 15 returns to `s`, hence the minimal non-trivial period is 15, and finally `not (15 ∣ 5)`. The axiom audit (`lean4/Check.lean`) reports only `propext` and `Quot.sound`, with no `sorryAx` and no `Lean.ofReduceBool`.

The Python script `reproduce.py` (standard library only) iterates (1) independently, prints the explicit 15-cycle, the period sets of Table 1, and the divisibility failures, and reports **PASS**.